

Abaqus jobs parallelization procedure in a SLURM cluster¹; 2nd edition

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¹With the PMTE-SC framework in mind

Preliminaries

- 1 The subprocess module must be imported into abaqus python; jobs in a batch must be launched with the non-blocking `subprocess.Popen()` command.

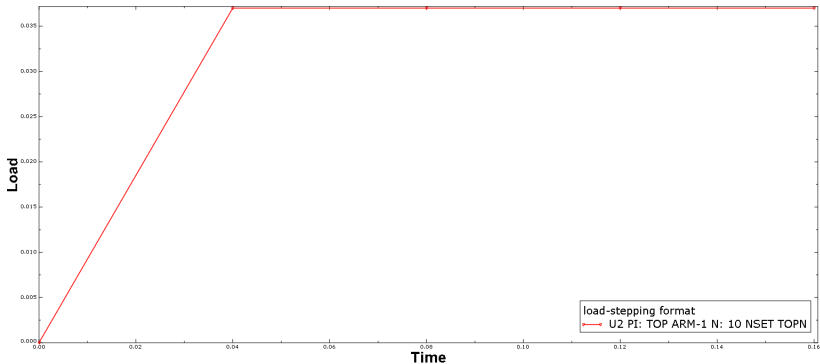
Preliminaries

- 1 The subprocess module must be imported into abaqus python; jobs in a batch must be launched with the non-blocking `subprocess.Popen()` command.
- 2 Whether the job parallelization technique is performed with Job Array Support² or by a master Python script, the number of concurrent jobs will be limited by the number of available Abaqus license tokens in the server.
- 3 Given the condition (2), an alternative strategy is to stage job execution into separate batches, e.g., two or more job groups, depending on the available license tokens.
- 4 Each batch job must (preferably) have its own directory to avoid interference with abaqus .lck and other files.

Filename	Filesize	Filetype
DCBuinter_L237m114_N10_K15_scratch		File folder
DCBuinter_L237m114_N11_K15_scratch		File folder
DCBuinter_L237m114_N1_K15_scratch		File folder
DCBuinter_L237m114_N2_K15_scratch		File folder
DCBuinter_L237m114_N3_K15_scratch		File folder
DCBuinter_L237m114_N4_K15_scratch		File folder
DCBuinter_L237m114_N5_K15_scratch		File folder
DCBuinter_L237m114_N6_K15_scratch		File folder
DCBuinter_L237m114_N7_K15_scratch		File folder
DCBuinter_L237m114_N8_K15_scratch		File folder

Preliminaries: load stepping

- Perhaps the single most important remark for the PMTE-SC to work properly with the subroutine coding of the minimization scheme is that the loading must be driven by a constant amplitude function.
 - Abaqus .inp file Amplitude definition:
*Amplitude, name=AMP-1
Initial step time, Value, Final step time, Value



Application

```
Joblist=[]
processlist=[]
# ***ABAQUS JOB PARALLELIZATION APPROACH:***
abaqus_executable = '/opt/abaqus/Commands/abq2022'
for job in range(1,len(Uinterprops)-4):
    # Build a list of command arguments
    commandN = [
        abaqus_executable,
        'job=' + myJobN.name,
        'user=' + nameSUBR,
        'interactive']

    # subprocess.Popen launches the command in a new process
    # and the script immediately continues to the next line.
    process = subprocess.Popen(commandN, cwd=work_dir)
    processlist.append(process)
```

- * 'interactive' keyword is essential for a post-processing free of errors.
- * The cwd argument ensures that each job is run in its working directory.

Wait for end of the jobs and post-process stages:

```
# --- LOOP 2: WAIT FOR JOBS TO FINISH ---  
    # This loop will pause the script until every job is done.  
    for i, proc in enumerate(processlist):  
        # .wait() method blocks execution until the process finishes  
        return_code = proc.wait()  
  
# --- LOOP 3: POST-PROCESSING ---  
    #  
    for job in Joblist:  
        # Open .odb files and extract simulation data.
```

Non-blocking vs blocking script approaches: i.e. stopping or not Python execution after a job submission

Coding feature	<code>subprocess.call()</code>	<code>subprocess.Popen()</code>
Behavior	Blocking; Python sleeps	Non-Blocking ³
Script complexity	Low	Medium; requires careful polling
Suitable for:	Crack propagation	Crack onset; parallel jobs

³Python immediately continues with script execution

Summary

PMTESC-3D computational efficiency pillars:

- Load increment bisection scheme: variable load increment magnitude.
- Job parallelization, e.g., different model parameters for the same applied load; this procedure allows concurrent jobs to run under the same cluster node⁴

```
process = subprocess.Popen(commandN, cwd=work_dir)
```
- j_{ama} iterations: convergence of damage criteria within the user subroutine file; either through the relative error of the potential energy $\Delta\P$ variation or changes in the damage array across step increments⁵, after which the fortran **CALL** XIT() command should be issued.

⁴This approach is most compatible with the subprocess module commands, as opposed to the job.submit() method of abaqus Python.

⁵Both criteria become relevant for large size models

Acknowledgements

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